

Highly Scalable and Reliable Video Streaming Platform

System Design

Highly Scalable and Reliable Video Streaming Platform
System Design

Generated on 2025-08-10 03:06 UTC

One Pager Summary

Goal: Deliver a global video on demand platform for discovery and playback of long and short form content with fast startup, smooth adaptive streaming, and privacy by design.

Users: Viewers on web, mobile, and TV; creators and studios uploading content; administrators and moderators; advertisers when ad supported tiers exist.

Core Value: Rapid play start, near zero rebuffering, accurate recommendations, and secure content delivery with DRM across devices.

North Star Metrics: Video start time under 2 seconds on median networks, rebuffer ratio under 0.2 percent, p95 API latency under 150 milliseconds, content availability 99.95 percent or better, data durability eleven nines for masters and mezzanines.

Scope: Catalog, ingestion, transcode pipeline, DRM protected delivery, playback, search and recommendations, watch history and continue watching, subtitles and audio tracks, analytics, moderation, and optional ads or subscriptions.

Functional Requirements

Browse catalog by genres, topics, channels, and personalized rows like because you watched and trending. Search with type ahead and filters. View video details including synopsis, cast, runtime, maturity rating, and available audio and subtitle tracks. Play video with adaptive bitrate, seek, captions, audio selection, and next up autoplay. Continue watching across devices. Track progress and watch history with privacy controls. Like, dislike, and save to lists. Optional user uploads with validation and moderation. Admin tools for content ingestion, policy, and entitlements. Monetization via subscriptions and or ad supported playback using server side ad insertion. Localization and accessibility across captions, audio description, and UI. Optional downloads to device for offline viewing with license duration controls.

Non

Availability 99.95 percent platform wide and 99.99 percent for CDN origin and manifest endpoints. Latency targets for p95 API reads under 150 milliseconds, catalog page assembly under 300 milliseconds, and video time to first frame under 2 seconds on median networks and devices. Consistency model is read your writes for playback entitlements and user progress, eventual consistency cross region under 2 to 5 seconds for catalog updates. Durability eleven nines for master assets and mezzanine files in multi region object storage. Scalability to tens of millions of daily active viewers, millions of concurrent streams, and billions of analytics events per day. Privacy and compliance aligned with SOC 2 type 2, GDPR, and CCPA with data minimization and deletion workflows. Cost efficiency via CDN offload, tiered storage, just in time packaging, and smart prewarming.

Architecture Overview

Client: Apps for web, iOS, Android, and TV platforms with a shared TypeScript or Kotlin Multiplatform SDK for API and playback control. Uses Media Source Extensions on web and native players on mobile and TV. Integrates with platform DRM.

Edge: CDN provides caching for manifests, media segments, images, and subtitles. Supports token validation, signed URLs, and image and text track compression.

Gateway: API Gateway with WAF and DDoS protection routes requests to microservices. REST and GraphQL are supported. Authentication with OAuth and OIDC using short lived tokens.

Core Services

Catalog Service stores titles, seasons, episodes, assets, media variants, images, ratings, and availability windows. Entitlement Service authorizes playback by plan and region and device count. Playback Service issues playback tokens and returns DRM and manifest locations.

Ingestion

Service accepts uploads and cloud imports with validation. Transcode Orchestrator manages encoding jobs. Packaging Service produces HLS and DASH manifests and segments on demand or ahead of time. DRM License Service integrates with Widevine, PlayReady, and FairPlay. Recommendation Service provides personalized rows and next up. Search Service indexes catalog. User Data Service stores lists, likes, and progress. Notification Service schedules release reminders. Ads Service handles server side ad insertion when enabled. Analytics Pipeline collects client and CDN events. Admin Service manages content, policies, and takedowns. Control Plane manages autoscaling, canaries, and configuration.

Data Flow

and Transcode

Studio or creator uploads a master file and metadata to object storage.

validates container and

codecs and writes an event to the log. Transcode Orchestrator schedules encoding to multiple bitrates and resolutions producing mezzanines. Packaging either performs just in time transmuxing to HLS and DASH or pre packages into segments and manifests with encryption keys assigned.

Playback

Client requests playback for a title. Gateway calls Entitlement to check plan, region, and device concurrency. Playback Service returns a signed manifest URL and DRM license endpoints. Player fetches manifest and segments from CDN. Adaptive bitrate selects an initial rendition using bandwidth estimate and device constraints and shifts up or down based on buffer health. Subtitles and audio tracks load on demand. Client emits playback events for analytics and recommendations.

Recommendations and Discovery

Home screen requests rows. Recommendation Service assembles rows with collaborative filtering, embeddings, and business rules. Catalog Service provides artwork and metadata. Client prefetches details for hover previews where supported.

Search

Client sends query to Search Service. Results are ranked by quality, freshness, and personalization signals with spell correction and synonyms.

Technology Stack

Client: TypeScript and React for web with a production player using Media Source Extensions and Encrypted Media Extensions; native players on iOS and Android; platform players on TV devices.

Edge and Delivery: CDN with support for tokenized URLs, signed cookies, HLS and DASH, low latency modes where needed, and HTTP or QUIC transport.

Gateway: Envoy or managed API Gateway and WAF. OAuth and OIDC with PKCE for apps and device code flow for TVs.

Messaging: Kafka or Kinesis for ingestion, packaging, and analytics events.

Compute: Kubernetes for microservices with horizontal autoscaling and KEDA triggers from queue lag.

Metadata Stores: Postgres for catalog and entitlements with read replicas and row level security for multi tenancy; Redis for caches and locks; DynamoDB or Cassandra for high write paths like progress tracking if needed.

Media Storage: Multi region object storage for masters, mezzanines, packaged segments, images, and subtitles.

Search: OpenSearch or Elastic managed service.

DRM: Widevine, PlayReady, and FairPlay license services behind token and device checks.

Recommendations: Feature store on Redis or DynamoDB, model serving on a managed platform, offline feature generation in a data lake.

Observability: OpenTelemetry, Prometheus, Grafana, centralized logs, and real user monitoring. IaC with Terraform and progressive delivery with canary and feature flags.

Playback and ABR Algorithms

Adaptive bitrate uses a hybrid strategy based on throughput and buffer aware heuristics. Startup selects a conservative bitrate based on network and device profile and quickly ramps using fast switching when buffer

is stable. Downshift thresholds are tuned to keep rebuffer ratio under target. Segment durations of four to six seconds balance CDN efficiency and interactivity. Low latency modes use shorter segments and partial segment delivery when required. Prewarm manifests and first segments on fetch to reduce startup. Trick play uses preview sprites at multiple resolutions. Caption rendering is offloaded to native platform when supported to reduce jank. Metrics include time to first frame, mean bitrate, rebuffer count, drift, and playback failure rate.

Data Model Sketch

Tenant: id, name, plan, region, branding, status, createdAt.

User: id, tenantId, profileId where households apply, plan, region, parental controls, settings, createdAt.

Title: id, tenantId, type movie or episode, seriesId optional, seasonId optional, metadata title and synopsis and cast and genres and maturity rating and release date, duration, availability windows, artwork and trailers, createdAt.

Asset: id, titleId, masterUri, mezzanineUris, packagedUris per protocol, drmPolicyId, imageUris, textTrackUris, status, createdAt.

PlaybackSession: id, userId, deviceId, titleId, startTs, endTs, lastPosition, bitrateStats, errors.

Entitlement: id, tenantId, planId, region rules, device concurrency, playback windows.

ListItem: id, userId, titleId, type favorite or watchlist, ts.

Interaction: id, userId, titleId, type like or dislike, ts, meta.

AdPolicy optional: id, tenantId, targeting rules, frequency caps, creative references.

Analytics event schema for play, pause, seek, rebuffer, and quality change with timestamps and metrics.

Storage Strategy

and Caching

Masters and mezzanines in multi region object storage with versioning and lifecycle policies. Manifests and segments are cached at CDN edge with high hit rates. Origin shield reduces origin load. Tokenized URLs and signed cookies control access. Catalog reads are cached in Redis and edge caches with short ttl. User progress uses write back cache with batch persistence to the database to absorb spikes. Precompute popular title rows and store in edge key value for fast home load. Subtitles compressed and cached with brotli. Images are transformed at the CDN with responsive sizes and modern formats.

Latency Budgets and Throughput

Startup budget targets two seconds to first frame. Client negotiation and token exchange under 200 milliseconds. Manifest fetch under 100 milliseconds from edge. First segment time to first byte under 150 milliseconds. API read p95 under 150 milliseconds. Write paths for progress and likes under 120 milliseconds. Throughput example ten million concurrent streams globally, one to two million manifest requests per second, tens of millions of segment requests per second, and billions of analytics events per day. Autoscale on queue lag, CPU, and request rates. Shed load gracefully on overload with client hints and reduced concurrency.

SLA and SLOs

Platform availability 99.95 percent monthly. Playback critical path and manifest endpoints 99.99 percent. Error budget 0.5 percent. RPO zero for ingestion logs and object storage replication. RTO per region under 15 minutes with automatic failover to another region. Player level fallbacks for CDN errors by switching to backup domains.

Security Privacy and Compliance

DRM enforced playback with Widevine, PlayReady, and FairPlay and device binding. Tokenized manifests and segment URLs with short expirations, signed with key management service. Forensics watermarking on high value titles. API least privilege scopes, mTLS for service to service, and secret rotation. DLP for uploads and studio deliveries to block leaks. Data minimization and pseudonymous analytics. GDPR and CCPA subject request tooling and configurable retention for watch history and analytics. SOC 2 controls for change management, access reviews, logging, and incident response. Vendor risk management and annual

penetration tests.

Multi Tenancy

Logical isolation by tenant id with row level security in catalog and user data. Per tenant branding, content rights, and pricing. Per tenant encryption keys and DRM policies. Regional data residency per tenant where required. Quotas and rate limits per tenant and contract tier. Dedicated VPC and regional isolation options for

enterprise customers.

Ads and Monetization Options

Subscription only avoids ads and simplifies playback. Ad supported tiers use server side ad insertion to avoid ad blockers and ensure smooth playback. Ad decisioning via third party ad servers with VAST and VPAID where appropriate. Frequency capping and brand safety policies applied. Payments integrate with app stores and web processors with entitlements updated in near real time.

CI CD and Deployment

Trunk based development with short lived branches. Automated tests and security scans in CI. Infrastructure as code with Terraform and environment workspaces. Progressive delivery with canary releases and shadow traffic for new playback and recommendation logic. Automatic rollback on SLO regression based on metrics and error budgets. Feature flags to gate new codec support or ABR policies per cohort.

Observability and Operations

End to end tracing across gateway, services, and data stores. Metrics include time to first frame, rebuffer ratio, average bitrate, playback failure rate, p50 p95 p99 latencies, cache hit rate, queue lag, and error budgets. Real user monitoring captures device, network, and player metrics. Logs are structured and scrubbed. Alerting on SLO burn rate, increased rebuffer, CDN error codes, license failures, and origin saturation. Dashboards per region and tenant with drill downs. Weekly operations review with action items.

Capacity Planning and Cost

Reference scale per region includes Kubernetes node pools for API and orchestration, three to five Kafka brokers with three day retention, Redis cluster with nine to twelve shards, Postgres primary with read replicas or DynamoDB tables sized for catalog and entitlements, search cluster with three data nodes and two masters, and object storage as origin with origin shield. CDN carries the majority of traffic. Monthly cost ranges from high five to mid six figures per region depending on concurrent streams, codec mix, and egress. Cost levers include CDN cache hit rate, segment duration, codec efficiency, tiered storage, and just in time packaging versus pre packaging trade offs.

Testing Strategy and Failure Drills

Unit tests for API, entitlement checks, and player helpers. Property tests for ABR policy and segment selection. Contract tests for license servers and CDN token validation. Load and soak tests for manifests and segments including burst scenarios for popular releases. Chaos tests for origin failover, CDN degradation, DRM license delays, and message queue stalls. Replay testing from production like logs in staging. Security testing with SAST, DAST, dependency checks, and secret scanning.

Phased Build Plan

MVP six to eight weeks

Catalog service, ingestion of masters, basic transcode ladder to H264, packaging to HLS, tokenized playback, captions, search basics, home rows, watch history, single region with multi AZ and CDN, metrics and dashboards.

Phase 2 two to three months

DASH and CMAF, multi DRM, improved ABR, downloads, recommendations and continue watching, multi AZ and canary deploys, analytics pipeline, and moderation.

Phase 3 three to six months

Multi region active active, advanced recommendations with embeddings, server side ad insertion for ad tiers, data residency, 4K and HDR codec ladders, offline and prefetch optimizations, and enterprise tenant isolation.

Key Design Choices Trade Offs and Risks

Just in time packaging reduces storage and increases CPU at peak while pre packaging increases storage but reduces peak compute. Choose hybrid by pre packaging for popular titles and JIT for tail content. Short segments reduce latency but increase overhead and CDN load; pick four to six seconds for general VOD and shorten only when needed. Multi DRM increases complexity but is required for broad device support. Recommendation models can introduce bias; add explainability and policy levers. Egress and media processing dominate cost; optimize with better codecs, higher CDN hit rates, and lifecycle policies.